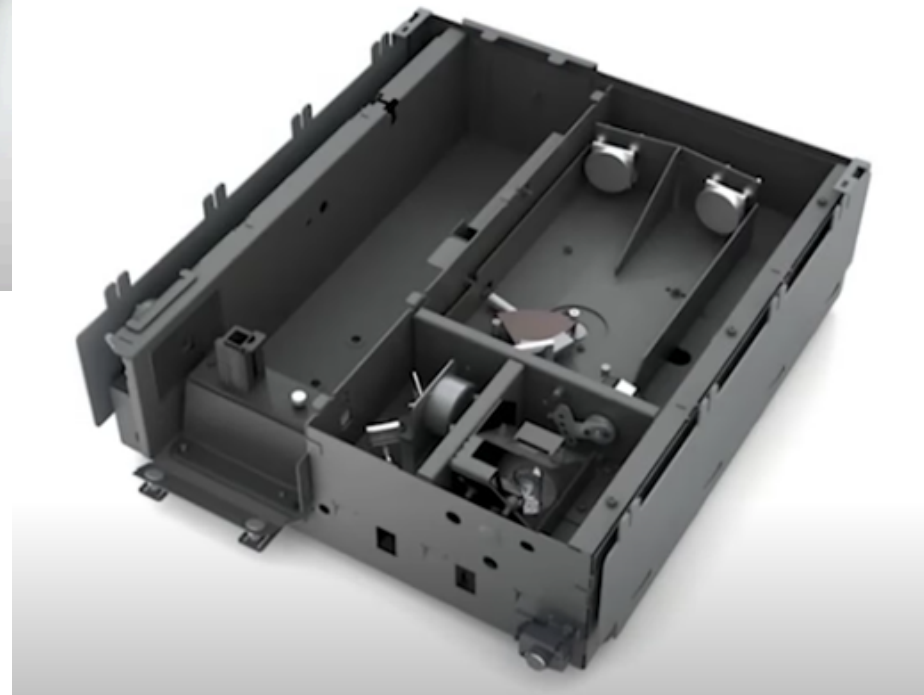
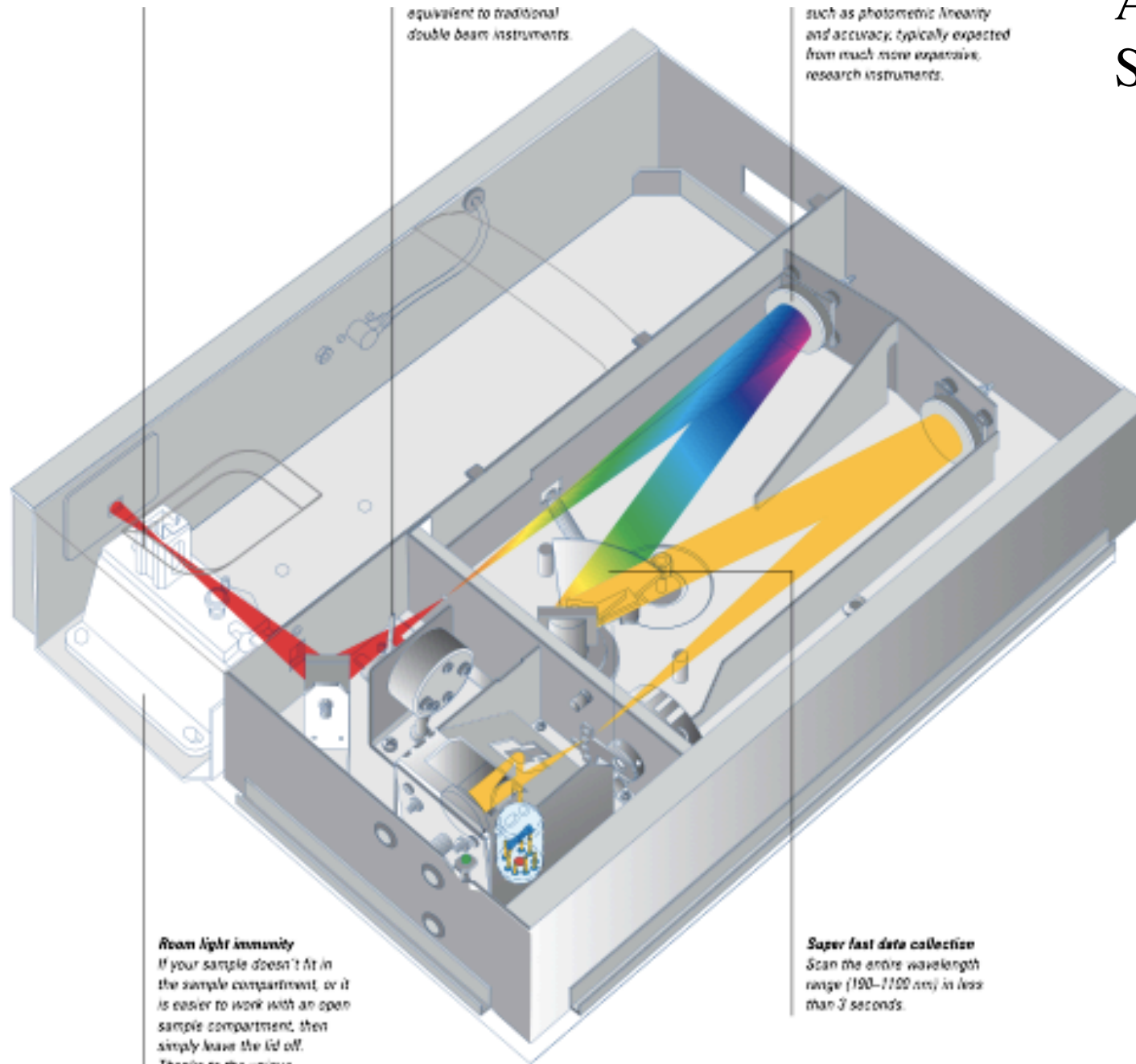


Agilent Cary60 Spectrophotometer



Agilent Cary60 Spectrophotometer



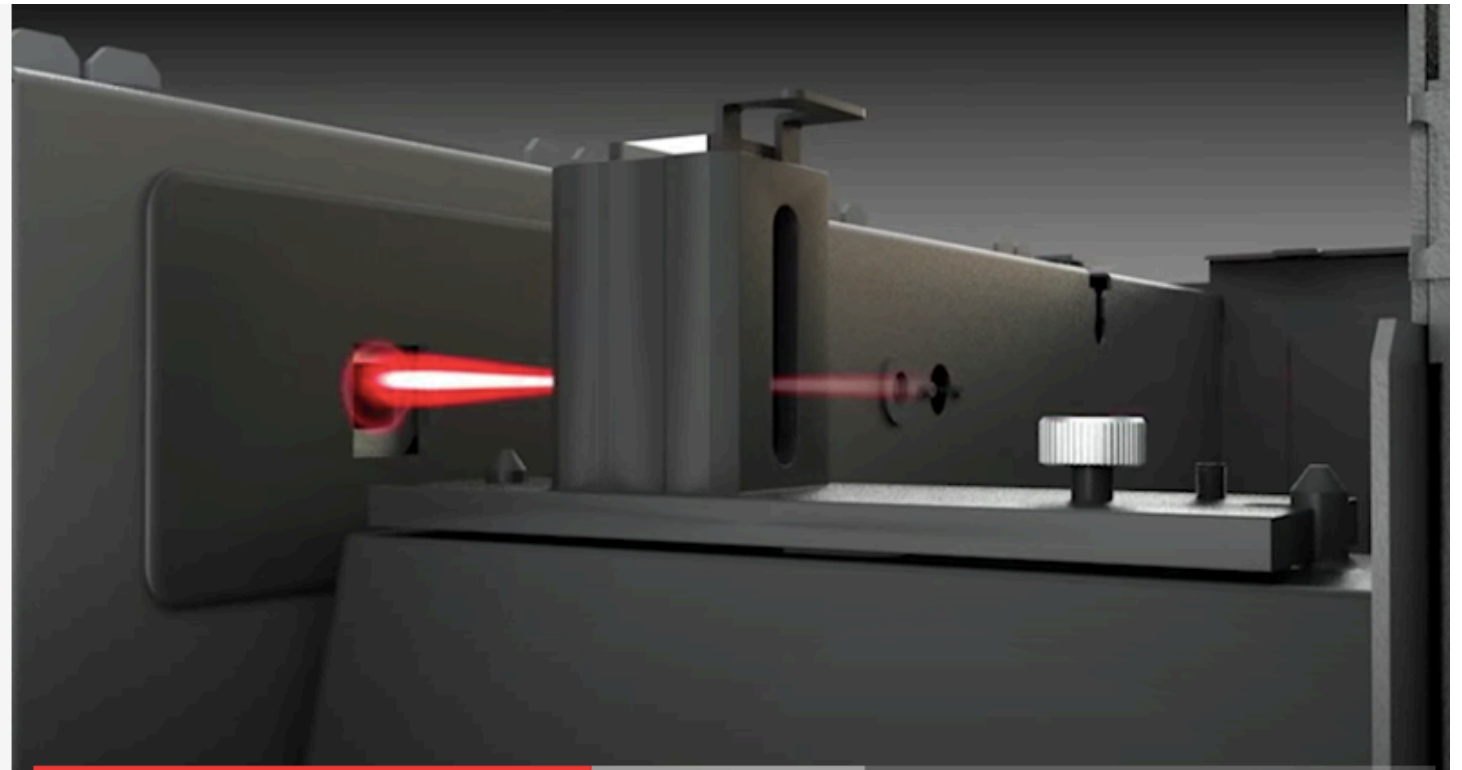
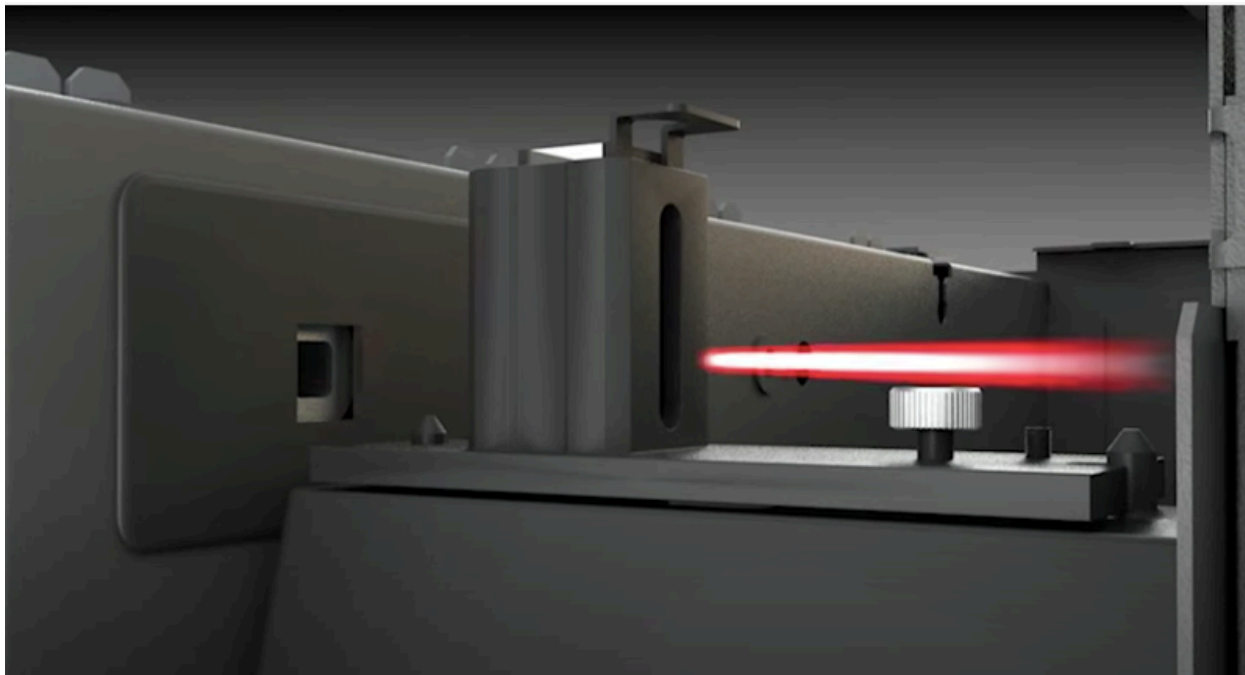
equivalent to traditional
double beam instruments.

such as photometric linearity
and accuracy typically expected
from much more expensive,
research instruments.

Room light immunity
If your sample doesn't fit in
the sample compartment, or it
is easier to work with an open
sample compartment, then
simply leave the lid off.
Thanks to the unique
properties of the Xenon flash
lamp, the Agilent Cary 60 is
not affected by room light.

Super fast data collection
Scan the entire wavelength
range (190–7100 nm) in less
than 3 seconds.

Agilent Cary60 Spectrophotometer



Overview

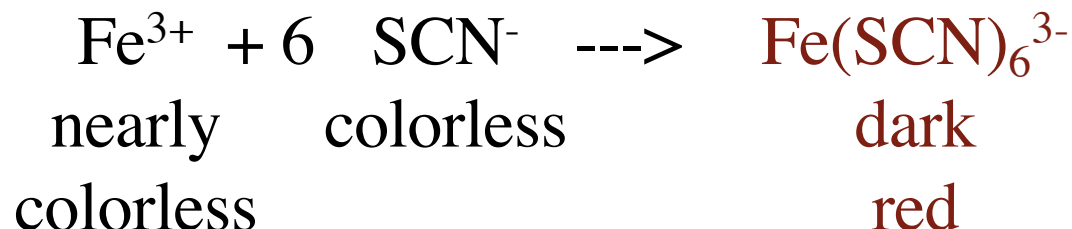
- Using color developing agents
- Absorption spectra and selection of optimum wavelength
- Limitations of Beer's law
- Best range of Absorbance for quantitative analysis

Beer's law in Chemical Analysis

- Compound must absorb light to be analyzed by Beer's law
- What if a sample does not absorb; i. e. $a = 0$?
Add a complexing agent to obtain a product that does!
- Less interferences if absorption is in the visible region

Beer's law in Chemical Analysis

Example:



Requirement of color developing reaction:

C_{product} is proportional to initial C_{analyte}

Properties of Good Color Developing Reagents:

- Large molar absorptivity for the product.
- Small molar absorptivity of the color developing reagent(s).
- Large K_{eq} for the formation reaction(s).
- Color developing reagent(s) that are much more soluble than the minimum required for complete conversion of analyte to product.
- Fast reaction.

Desirable conditions for quantitative analysis using Beer's law

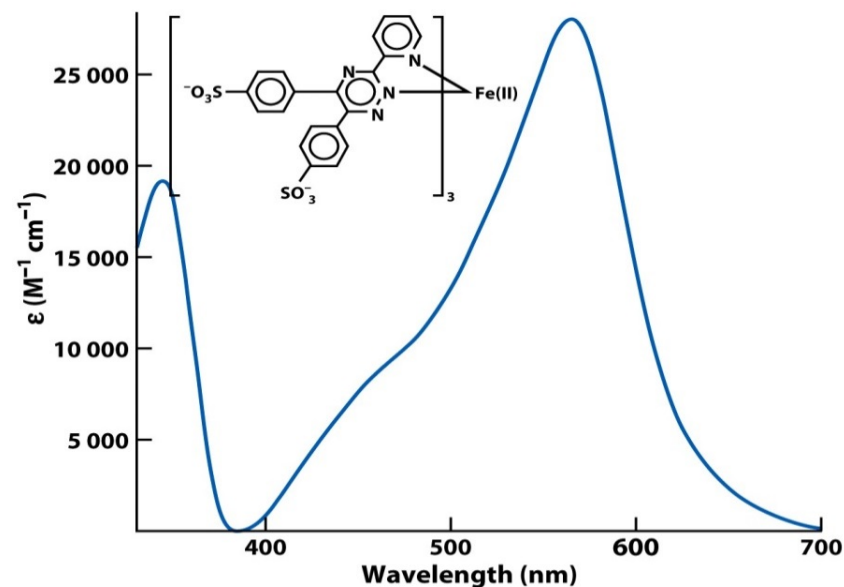
High sensitivity	<i>large slope (slope = ab)</i>
Wide linear range	<i>limitations of Beer's law</i>
Reduced error from calibration curve	• <i>large slope</i> • <i>optimum Absorbance range</i>

$$s_c = \frac{s_r}{|m|} \sqrt{\frac{1}{M} + \frac{1}{N} + \frac{(\bar{y}_u - \bar{y})^2}{m^2 \sum (x_i - \bar{x})^2}}$$

Selecting the wavelength - Absorption Spectrum

Spectrum:

A plot of T , % T , A or absorptivity vs λ , ν , E or any variables proportional to any of the variables used to characterize em radiation.



Absorption spectrum

$$A(\lambda) = a(\lambda) b C$$

Absorptivity, a (or **molar** absorptivity, ϵ):

- An intrinsic property of a chemical species.
- A measure of how strongly a chemical species absorbs light at a given wavelength.

$$\epsilon = 8.7 \times 10^{19} P A$$

where P is electron transition probability (0-1) and A is the capture cross section area of the molecule ($\sim 10^{-15} \text{ cm}^2/\text{molecule}$)

Wave properties

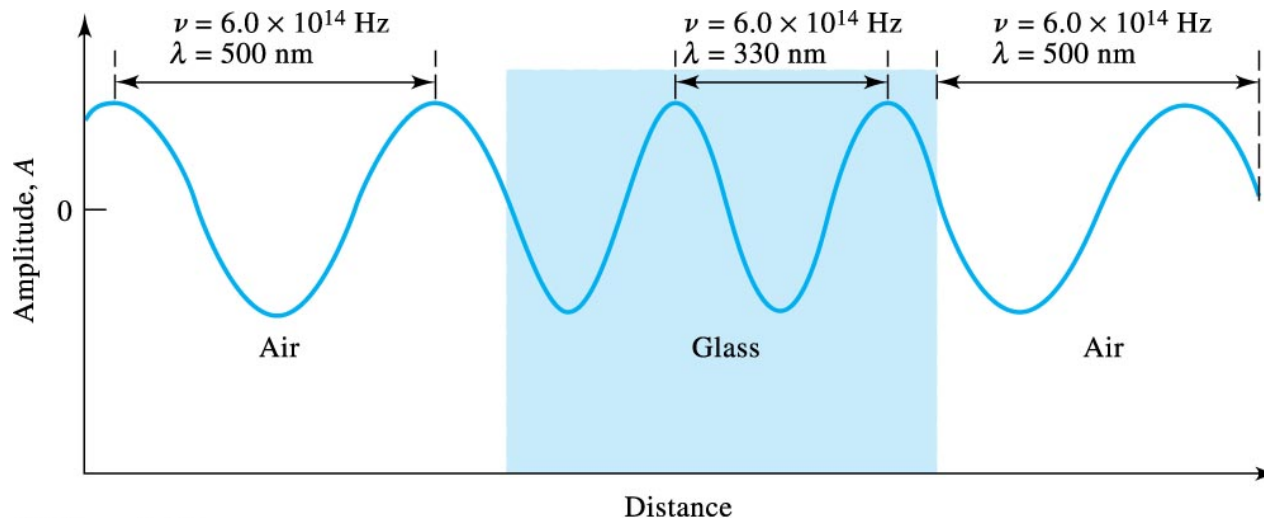


Fig 24-2

© 2004 Thomson - Brooks/Cole

ν (frequency): Depends on property of emitter

Never changes when propagating in different media

λ : Depends on property of medium & wave.

- Speed=distance /time, $c = \lambda * \nu$
- In vacuum, $c = 2.998 \times 10^8$ m/s.
In any other medium, speed of light = c/n where n = refractive index

Limitations to Beer's law

Beer's law is applicable mainly at *low conc.*

At high concentrations:

- (1) interactions between particles alters charge distribution either in the absorbing species or excited species or both and thus changes the energy needed for excitation
- (2) refractive index of the medium may change.

Other conditions that might cause Beer's law to 'apparently' fail, even under low concentrations:

- Chemical
- Instrumental

Limitations to Beer's law

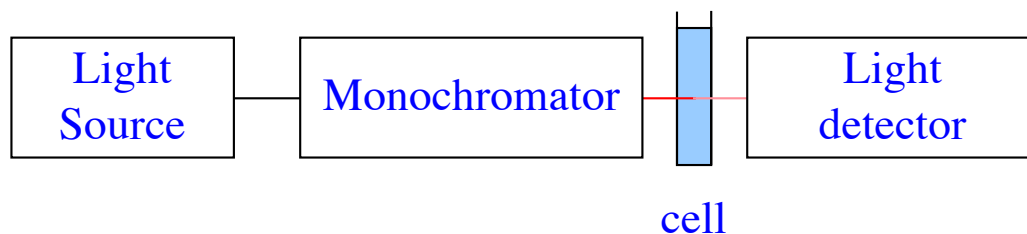
Chemical:

- Interactions between absorbing species that will affect their charge distributions
- Association, dissociation or reaction of the absorbing species with the solvent to generate a product that has a different absorption spectrum from analyte
- Incomplete reaction with color developing reagent: K_{eq} is too small, or too little color developing reagent
- Photodecomposition of the sample

Instrumental:

- non-monochromatic (polychromatic) light
- stray light

Apparent instrumental deviations from Beer's law

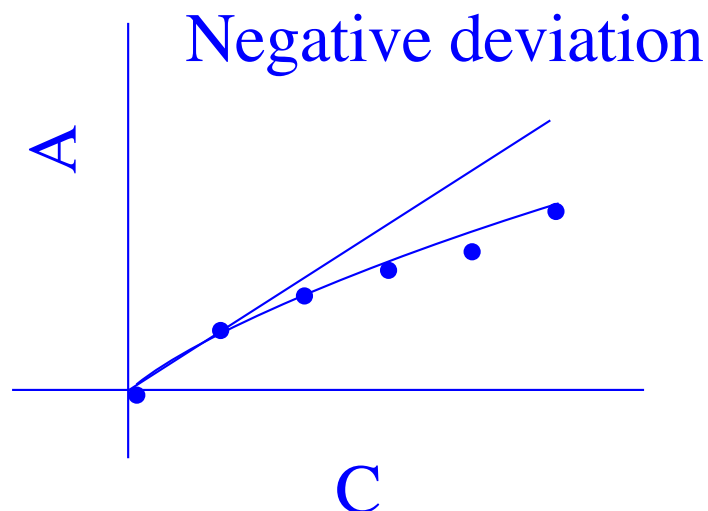


$$A(\lambda) = a(\lambda)bc$$

Monochromatic:

- Literally: one color, i.e. one wavelength.
- Spectroscopy: range of λ s reaching the detector is narrow compared to the range of λ s absorbed by the sample.

Anything that causes more light to reach the detector than Beer's law predicts causes a negative deviation from Beer's Law.



Apparent instrumental deviation - Polychromatic light

Effect of Two Wavelengths

$$A(\lambda) = a(\lambda)bc \qquad A = -\log \frac{P}{P_0} \qquad \frac{P}{P_0} = 10^{-abc}$$

$$\text{at } \lambda_1, P(1) = P_0(1) 10^{-a_1 b C} \qquad P_{\text{meas}} = P(1) + P(2)$$

$$\text{at } \lambda_2, P(2) = P_0(2) 10^{-a_2 b C} \qquad P_{0,\text{meas}} = P_0(1) + P_0(2)$$

$$T_{\text{meas}} = \frac{P_{\text{meas}}}{P_{0,\text{meas}}} = \frac{P(1) + P(2)}{P_0(1) + P_0(2)} = \frac{P_0(1) 10^{-a_1 b C} + P_0(2) 10^{-a_2 b C}}{P_0(1) + P_0(2)}$$

$$\text{When } a_1 = a_2 \quad T_{\text{meas}} = \frac{[P_0(1) + P_0(2)] 10^{-abC}}{[P_0(1) + P_0(2)]} = 10^{-abC} \quad \text{LINEAR A vs C}$$

$$\text{When } a_1 \neq a_2 \quad \text{NON-LINEAR A vs C}$$

Apparent instrumental deviations from Beer's Law

Question: If 1 % too much light reaches the detector, does it make a difference if A_{expected} is small (e.g. use 0.1) or large (e.g. use 1.0)? (i.e. at what end is the error larger?)

Consider:

$$\text{Case 1: } A_{\text{expected}} = 0.1$$

$$T_{\text{expected}} = 10^{-0.1} = 0.794...$$

$$T_{\text{measured}} = 0.804...$$

$$A_{\text{measured}} = -\log(0.804...) = 0.09456...$$

$$A_{\text{measured}} = 0.09456...$$

$$\text{Error} = -0.005433...$$

$$\text{Case 2: } A_{\text{expected}} = 1.0$$

$$T_{\text{expected}} = 10^{-1} = 0.10$$

$$T_{\text{measured}} = 0.11$$

$$A_{\text{measured}} = -\log(0.11) = 0.9586...$$

$$A_{\text{measured}} = 0.9586...$$

$$\text{Error} = -0.04139...$$

Conclusions: Errors in A are negative for non monochromatic light.

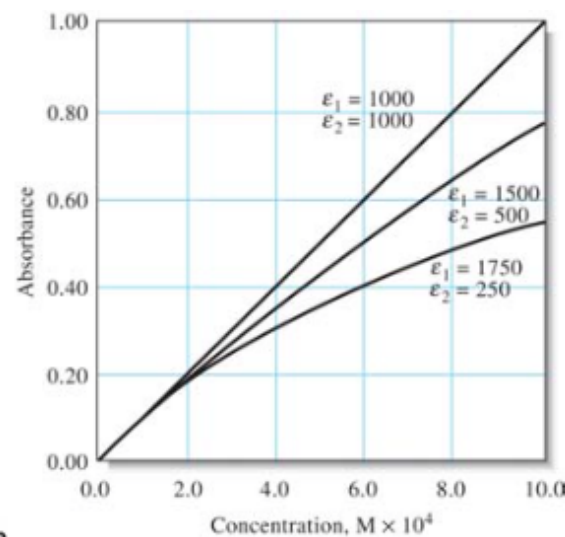
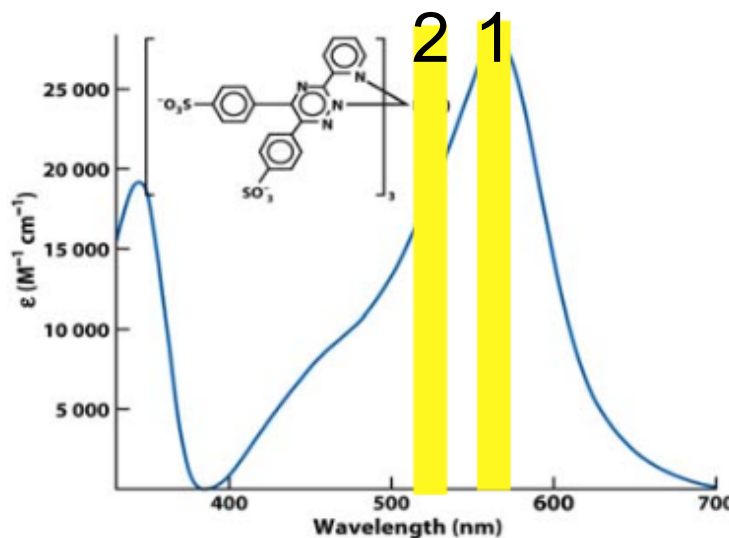
Errors in A are larger the bigger A_{expected} ; i. e. for larger C.

Would you predict that measurements of absorbances from a cloudy sample might lead to positive or negative deviations from Beer's law? (note: suspended particles cause light scattering)

Apparent instrumental deviation - Polychromatic light

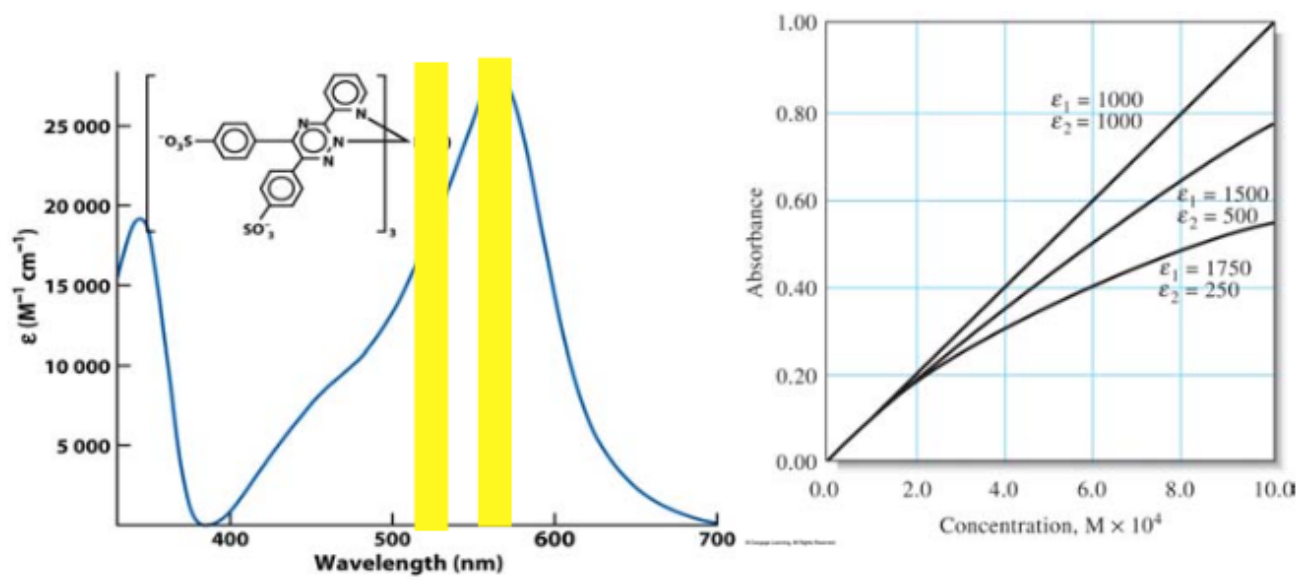
- Select a wavelength band (e.g. band 1) which corresponds to a region of the absorption spectrum in which absorptivity changes little with wavelength to avoid negative deviation from Beer's law
- When measurements must be made on a slope, linearity is improved by using a narrower effective bandwidth, a very narrow band 2 should be used for light to be considered monochromatic otherwise Beer's law will be obeyed over a narrower conc. range

Negative deviations
more pronounced at
high concentrations



Apparent instrumental deviation - Polychromatic light

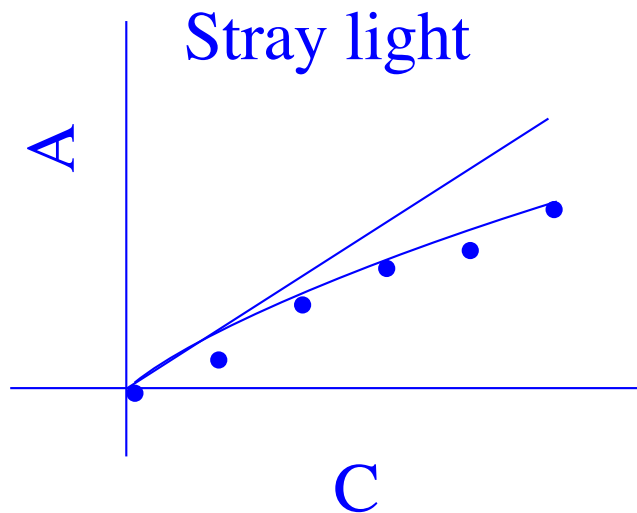
- Selecting a wavelength band at the top of an absorption peak ($\lambda_{\max}$) is also advantageous because it leads to increased slope and hence increased sensitivity and decreased error from straight line calibration!
- It is not always possible to detect at $\lambda_{\max}$, either because of spectral interference or because of limit on value of absorbance (A should be less than 1-1.5, more on that later)



Apparent instrumental deviation - stray light

Stray light: Light that reaches the detector without passing through the sample

$$A = -\log \frac{P + P_s}{P_0 + P_s}$$



- At low concentration of analyte, P_s is significantly smaller than P_0 and P , and the absorbance is unaffected by the stray radiation.
- At a higher concentration of analyte, less light passes through the sample and P and P_{stray} may be similar in magnitude. This results in an absorbance that is smaller than expected, and a negative deviation from Beer's law.

Error from stray light is most serious at high absorbances

Example (Harris p. 432)

If the true absorbance of a sample is 2.00 and there is 1.0% stray light. Find the apparent absorbance.

True transmittance $(P/P_0) = 10^{-A} = 0.010$

1% stray light $P_s = 0.01$

Apparent transmittance =

$(P+P_s)/(P_0+P_s)=$

$(0.01+0.01)/(1+0.01) = 0.0198$

The apparent absorbance

$A = -\log T = 1.70$ (instead of 2.00)

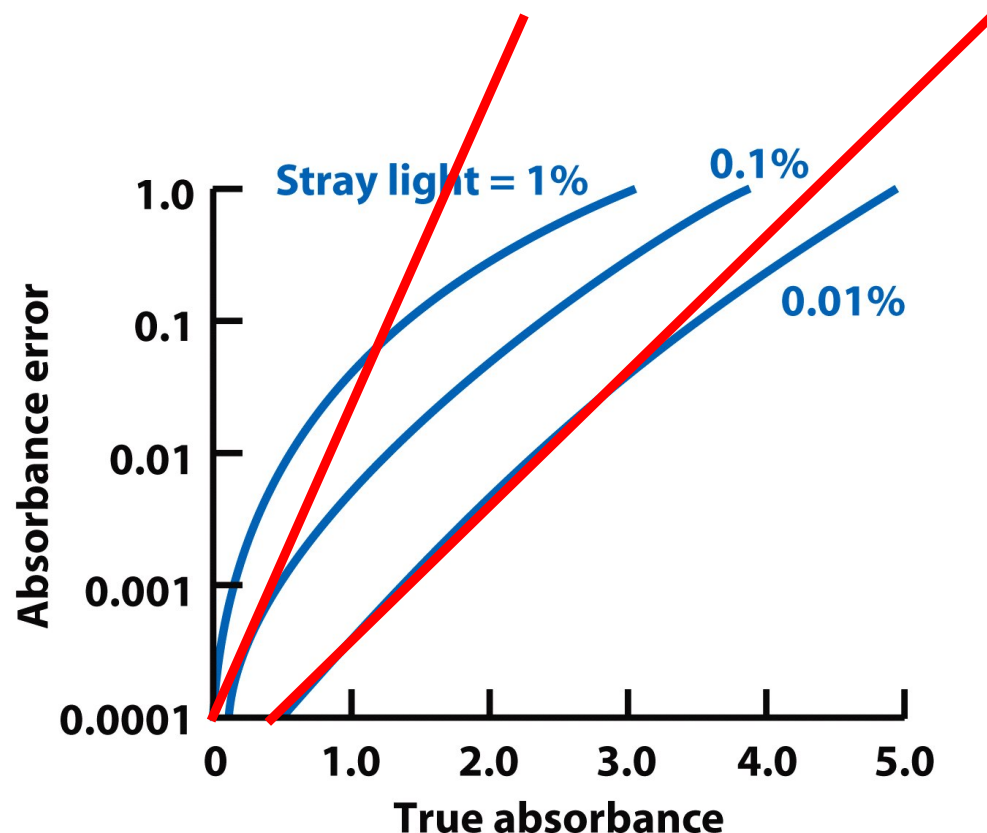


Figure 20-9
Quantitative Chemical Analysis, Seventh Edition
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Figure 20-9

What is the best range of A?

(26A-3 in your textbook - effect of instrumental uncertainty)

The uncertainty in a photometric concentration measurement varies in a complex way with the magnitude of the transmittance

$$A = -\log T = abC \qquad C = \frac{-1}{ab} \log T$$

$$\frac{\partial c}{c} = \frac{0.434}{\log T} \left(\frac{\partial T}{T} \right)$$

The best range of T (and therefore the best range of A) is one that minimizes error in the calculated concentration by Beer's law

Figure 26-10 and 26-11: for most spectrophotometers it is best to use solution concentrations such that A value are between 0.1 and 1 (2 for high end spectrophotometers)

What is the best range of A?

Suppose that the expected transmittance were T and the instrument measured $T + 0.001$, what would be the % relative error in C?

$$\% \text{ Er} = \left| \frac{C_{\text{obs}} - C}{C} \right| * 100 \%$$

$$= \left| \frac{A_{\text{obs}} - A}{A} \right| * 100\%$$

$$C = \frac{A}{a b}$$

$$T = 10^{-A}$$

$$A_{\text{obs}} = -\log(10^{-A} + 0.001)$$

$$C_{\text{obs}} = \frac{A_{\text{obs}}}{a b}$$

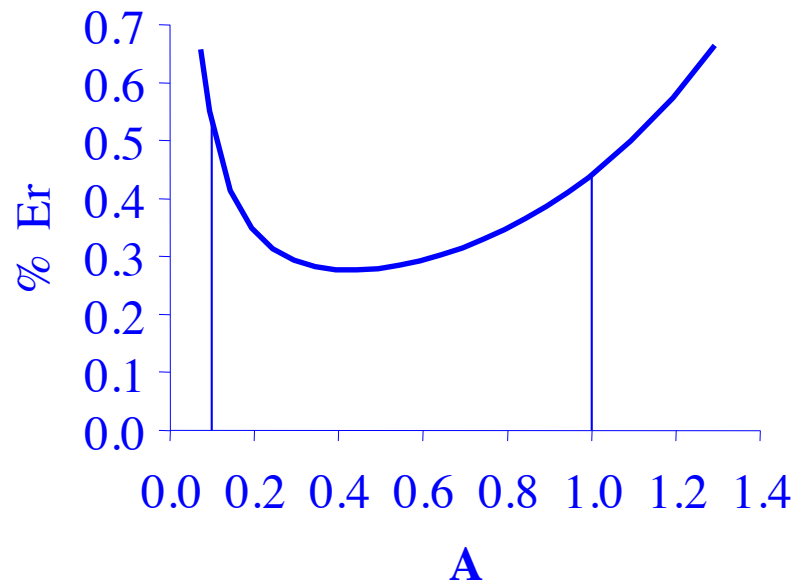
$$0.1 < A_{\text{opt}} < 1$$

$$\sim 80\% > \% T_{\text{opt}} > 10 \%$$

$$\frac{0.1}{a b} < C_{\text{opt}} < \frac{1}{a b}$$

$$m = a b$$

$$\frac{0.1}{m} < C_{\text{opt}} < \frac{1}{m}$$



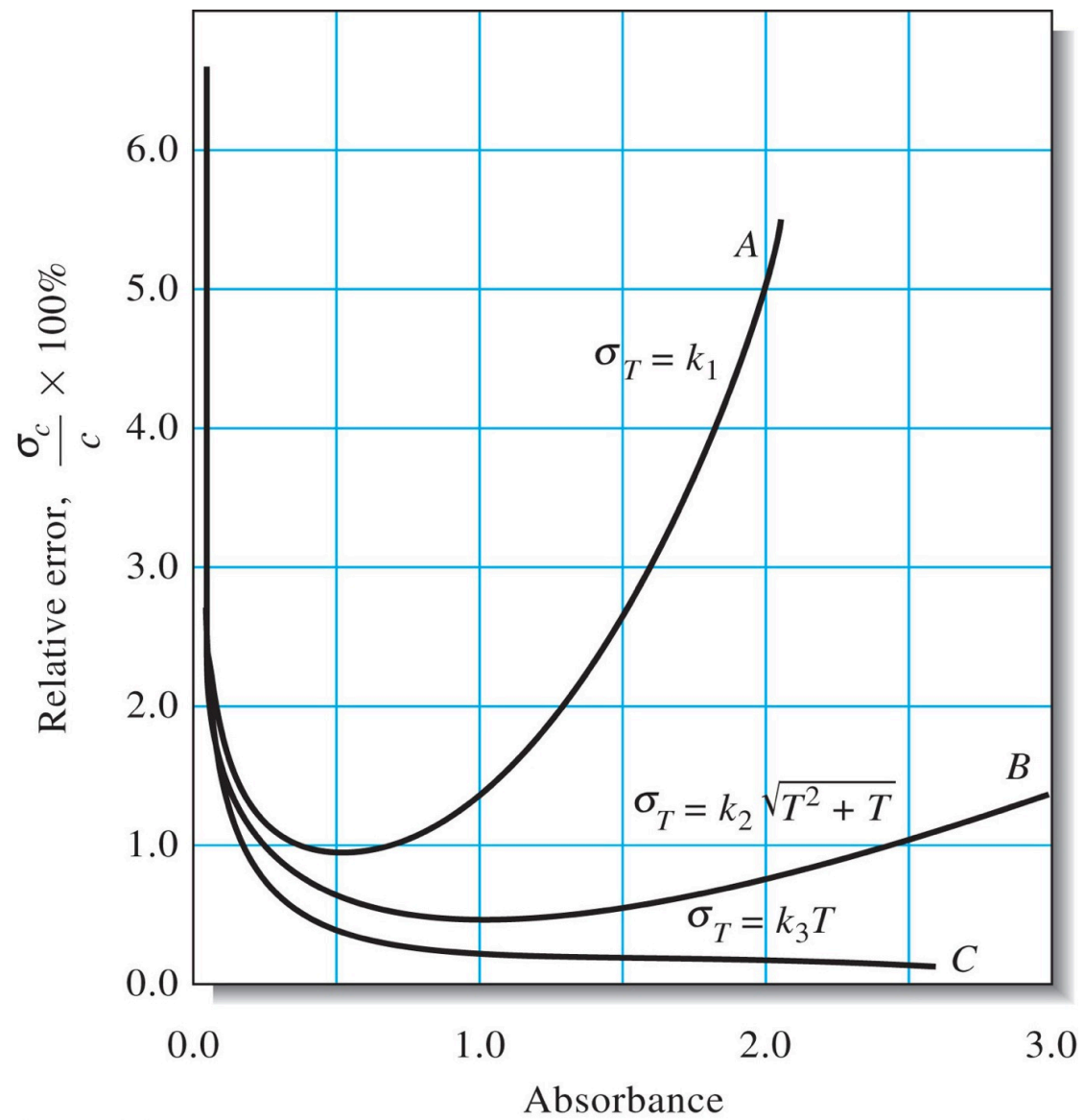


Figure 26-10 p737

There is an optimum range of A to target:

If Absorbance is too low (i.e. high Transmittance thru sample), the difference between P and P_0 is too small for accurate measurement.

If Absorbance is too high, too little light reaches the detector for accurate measurement.

Quantitative aspects of absorption measurements

Use [0.1 -1.0] as the optimum Absorbance range.

1. A certain analyte has a molar absorptivity of 9×10^4 . What would be the optimum range of concentrations for the set of standards prepared for analysis by absorption spectrophotometry in a 1.00-cm cell?

Beer's law single point calibration – Practice example 1

A 4.629 g sample of an alloy was dissolved in nitric acid; the resulting solution was transferred to a 100-mL volumetric flask and diluted to volume with water.

A 5.0 mL portion of the solution was pipetted into a 50 mL volumetric flask and diluted to volume with water. This solution was analyzed at 324.7 nm (Cu absorption line) by flame atomic absorption; a net absorbance of 0.119 was observed. A 2.00 mM standard Cu solution yielded a net absorbance of 0.125 on the same AA instrument.

Determine the %Cu in the alloy.

Beer's law single point calibration – Practice example 2

(Adapted from Harris 5th Ed. 19-13) A 10-mL aliquot of a $6.80 \times 10^{-4} \text{ M}$ Fe^{3+} was pipetted into a 50-mL volumetric flask, acidified with nitric acid and treated with excess KSCN to form a red complex with SCN^- ion before diluting to volume with deionized water. The resulting solution transmitted 39.1 % of the light in a 1.00 cm cell at 447 nm, which is a wavelength near the absorption peak for the complex.

A 25.0 mL sample of runoff water was acidified with nitric acid and treated with excess KSCN to form a red complex. The solution was diluted to 100.0 mL in a volumetric flask. The sample was found to transmit 67.8 % of the light in a 1.00 cm cell at the same wavelength. Calculate the concentration of Fe^{3+} in the runoff water.

recommended eoc 24 exercises:

3, 5, 9, 13, 14, 15, 19, 21, 23, 25, 26, 27, 28

will post Excel Solutions